

Functional & Performance Testing Template

Model Performance Test

Date	21 May 2026
Team ID	HDI Predictor Team
Project Name	HDI Predictor using Machine Learning and Flask
Maximum Marks	4 Marks

Test Scenarios & Results

Test Case ID	Scenario (What to test)	Test Steps (How to test)	Expected Result	Actual Result	Pass/Fail
FT-01	Numeric Input Validation	Enter valid and invalid indicator values in the prediction form.	Valid numeric inputs are accepted and invalid inputs are rejected.	Passed	
FT-02	Country Auto-Fill Validation	Select a country from the dropdown and confirm the fields populate.	Selected country values are filled correctly from the cleaned dataset.	Passed	
FT-03	Single Prediction Output	Submit one complete record and review the response page.	HDI score, category, confidence, and charts are displayed.	Passed	
FT-04	CSV Batch Prediction	Upload a CSV file with multiple countries.	The system returns a structured prediction table.	Passed	
PT-01	Response Time Test	Measure the time taken for a single prediction request.	The response should be quick and usable in the browser.	Passed	
PT-02	Batch Processing Test	Upload multiple records in a CSV file.	The system should process all rows without crashing.	Passed	

PT-03	Model Loading Test	Restart the Flask application and check model loading.	HDI.pkl loads successfully during startup.	Passed	
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